

Biodegradable polyurethane based clay composite and their anti-biofouling properties

Abid Ali^a, Yue Xiao^b, Lina Song^a, Jiankun Hu^a, Qingqing Rao^a, Muhammad Shoaib^a, Bilal Ul Amin^a, Xiaoli Zhan^{a,c,*}, Qinghua Zhang^{a,c,*}

^a College of Chemical and Biological Engineering, Zhejiang Provincial Key Laboratory of Advanced Chemical Engineering Manufacture Technology, Zhejiang University, Hangzhou 310027, PR China

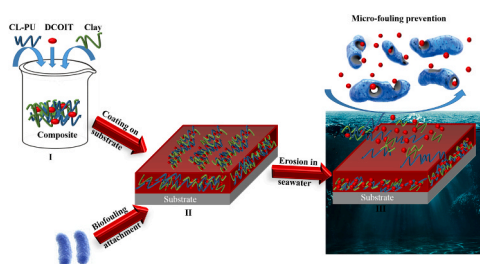
^b School of Mechanical and Electrical Engineering, Nanchang Institute of Technology, Jiangxi 330099, PR China

^c Institute of Zhejiang University-Quzhou, Quzhou 324000, PR China

HIGHLIGHTS

- Biodegradable composite synthesized by mixing of the solution method.
- The clay reduced the crystallinity and spherulite size of the PCL in the composite.
- The environmentally-friendly composite prevents the early stage of biofouling.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Clay
4, 5-dichloro-2-octyl-isothiazolone
Caprolactone
Biodegradable polyurethane
Anti-fouling coating

ABSTRACT

The colonization of tiny marine organisms on ship's hull which are connected directly from seawater, is a costly issue. Biofouling contributes to enhanced hydrodynamic drag, which leads to higher consumption of fuel and greenhouse gas emissions. Therefore, the environment-friendly solution is very necessary for anti-biofouling activity. For this purpose, we have synthesized biodegradable polyurethane (CL-PU) composed of ϵ -CL and 4,4'-methylenebis(cyclohexyl isocyanate) (H_{12} MDI) and 1,4 butanediol (1,4 BD). The synthesized CL-PU was further modified with 4, 5-dichloro-2-octyl-isothiazolone (DCOIT) and clay by mixing of the solution to make composites. Our study showed that CL-PU/DCOIT/clay composite degraded in the artificial seawater (ASW), enzymatic solution and seawater (through laboratory test) due to the contents of clay. Because the existence of clay reduced the size of the spherulite of polycaprolactone (PCL) in the composite, therefore remarkably improved the crystallinity as determined via the polarizing optical microscope (POM) and differential scanning calorimetry (DSC). The composite acted as carrier of antifoulant (DCOIT) and controlled their release rate. The anti-bacterial and anti-diatom experiments exhibited that the composite with DCOIT contents were effective in preventing the accretion of *Escherichia coli* (*E.coli*) (region inhibition 14 mm) and marine diatom *Navicula incerta* (82.5% reduction). The degradable CL-PU/DCOIT/clay composite will have tremendous durability and excellent anti-fouling activity for marine biofouling owing to sustainability and tunability without being toxic to aquatic biota.

* Corresponding authors at: College of Chemical and Biological Engineering, Zhejiang Provincial Key Laboratory of Advanced Chemical Engineering Manufacture Technology, Zhejiang University, Hangzhou 310027, PR China

E-mail addresses: xlzhan@zju.edu.cn (X. Zhan), qhzhang@zju.edu.cn (Q. Zhang).

<https://doi.org/10.1016/j.colsurfa.2021.126946>

Received 25 March 2021; Received in revised form 25 May 2021; Accepted 30 May 2021

Available online 1 June 2021

0927-7757/© 2021 Elsevier B.V. All rights reserved.

1. Introduction

The accretion of micro-organism onto the surface normally results in biofouling that causes a severe problem in the shipping industry, such as surface roughness, fuel consumption and corrosion rate, etc [1–3]. Marine biofouling is a worldwide issue, going to cost billions of dollars every year in transportation [4]. The creation of an efficient eco-friendly fouling-resistant coating is a major challenge.

In the old days most of the coatings were biocide-based self-polishing coatings and their major ingredient was tributyltin (TBT) as a versatile and effective biocide [5]. However, TBT was banned in 2008 due to its harmful effect on the environment and humans [6,7]. Although, Isothiazolone biocides are efficient and robust in microbial (bacteria, algae and fungi) control for different biomedical applications [8–11]. These biocides are efficient at low concentration, resilient and fast-acting in inhibiting metabolism and growth of microorganisms, besides being able to control the development of biofilm [12]. Isothiazolone biocides act as an electrophile agent which interacts with crucial enzymes to suppress metabolism and growth, with cell death arising after several hours of contact [13]. DCOIT is a kind of this biocide, that accomplishes its anti-microbial activity by taking action with thiol-containing enzymes [14]. DCOIT is an anti-fouling agent commonly used in marine anti-fouling applications as an alternative TBT biocide. DCOIT have shown better environmental behaviour compared to TBT and more effective inhibition of microorganism [13].

On the other hand, biodegradable polymers have an anti-fouling capability as their erosion or decomposition by the enzyme in the aquatic environment, which contributes to the self-renewing surface and can inhibit biofilm formation from growing and settling [15–17].

Nowadays, many efforts were made for using biodegradable polyester to develop an eco-friendly anti-fouling coating. For example, Fay et al., prepared L-lactide and D-valerolactone introduced into polycaprolactone to reduce crystallinity and their study demonstrated that the releasing of biocides had been strongly linked only to the degradation of polymers [18].

Ma et al., prepared poly (caprolactone-co-glycolide) PU coatings and managed to keep anti-fouling rise to 150 days with the introduction of DCOIT eco-friendly antifoulant. The mixture of the release of antifoulants and degradation of PU could significantly prevent the formation of marine biofouling [19].

Ma et al., also prepared PCL/clay coating to reduce the size of the spherulite by increasing the amorphous area and enhancing the water penetration of the composite. However, the DCOIT was introduced into the PCL/clay composite as an anti-fouling agent, thus it carried the DCOIT and sustain released into the seawater for 10 months and exhibited anti-fouling activity [17]. As a result, the combination of CL-PU with antifoulant (DCOIT) and clay is expected to demonstrate improved mechanical efficiency, degradation rate and eco-friendly anti-fouling performance in a complicated marine ecosystem.

We prepared CL-PU by open ring polymerization and condensation

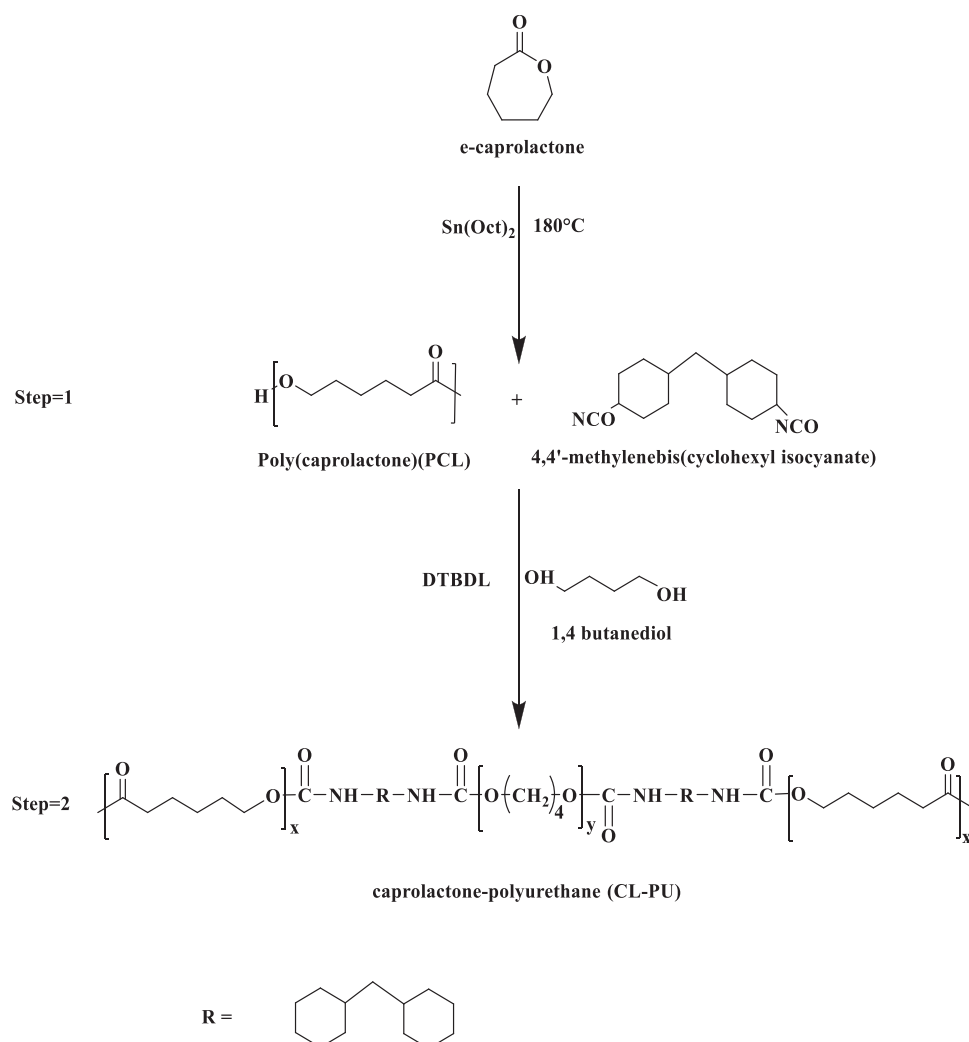


Fig. 1. The synthesis reaction of CL-PU.

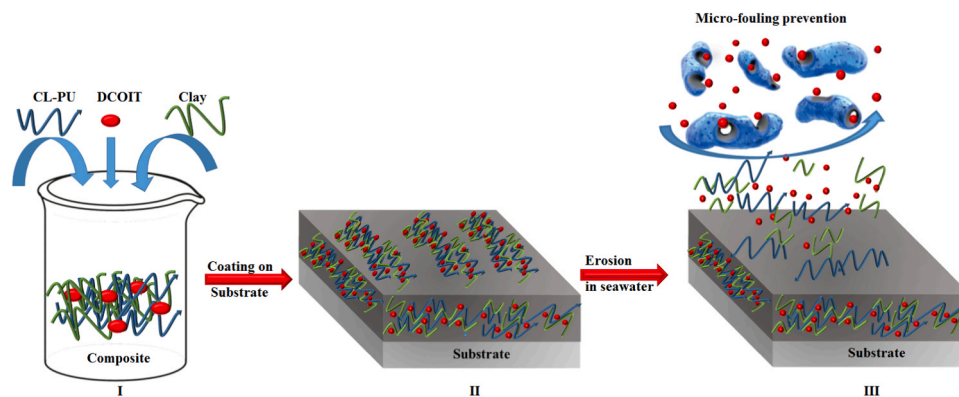


Fig. 2. (I) Schematic illustration for the synthesis of CL-PU/DCOIT/clay composite, (II) the coating casting on the substrate and (III) the release of DCOIT from composite during degradation and show preventing adhesion of the biofouling and biofilm formation.

reactions, then prepared CL-PU further modified with DCOIT and clay by mixing solution to make the composite. Composite was used as an organic antifoulant carrier and controlled release system. Our study has shown that the addition of clay speeded up the degradation rate of composite and enhanced the amorphous interfacial region by reducing the spherulite size and crystallinity of PCL. The adhesion strength of the composite was also enhanced by clay. The wettability, degradation and anti-fouling efficiency of the composite surface have been investigated. The prepared eco-friendly composite is provided to durability, non-toxicity and has promising properties to be used as an outstanding marine anti-fouling coating.

2. Experimental section

2.1. Materials

ϵ -caprolactone (99%, Aladdin), 4,4'-methylenebis(cyclohexyl isocyanate) (H_{12} MDI) (90%, Merck), 1,4 butanediol (99%, Aladdin), anhydrous toluene (99.5%, Aladdin), tetrahydrofuran (THF) (99%, Aladdin), dichloromethane (CH_2Cl_2) (99.5%, Aladdin), n-hexane (97%, Aladdin), anhydrous methanol (99.9%, Aladdin), dibutyltin dilaurate (DBTDL) (97%, Aladdin), Lipase (Aladdin), clay (Kaolin ($Al_2Si_2O_5(OH)_4$) (Shanghai Yuefang Industry & Trade Development Co) and 4,5-dichloro-2-octyl-isothiazolone (DCOIT) (98.0%, Macklin). The monomers were used without further purification. Stannous octoate ($Sn(Oct)_2$) (95%, Macklin). The artificial seawater (ASW) (pH 8.2) was prepared by NaCl, $MgCl_2 \cdot 6H_2O$, KBr, Na_2SO_4 , $NaHCO_3$, $CaCl_2$, KCl, H_3BO_3 , NaF, and $SrCl_2 \cdot 6H_2O$, while pH adjusted by HCl and 1 M NaOH [21]. We obtained natural seawater from Lincheng, Zhoushan Island, Zhejiang province, China.

2.2. Synthesis of polycaprolactone (PCL)

The PCL polymer was synthesized with the $Sn(Oct)_2$ catalytic converter by ring open polymerization process (Fig. 1) [22]. In the process, ϵ -CL (5 g, 45 mmol) and $Sn(Oct)_2$ (0.7% of total reactant) were diluted in 40 ml anhydrous toluene under a constant stirring. The mixture was deoxygenated with nitrogen for 15 min at room temperature and then reacted at 180 °C for 3.5 h (h) to obtain the raw product. Subsequently, the product was dissolved in CH_2Cl_2 and purified by precipitation with anhydrous methanol, then filtered and dried under vacuum at 40 °C for 24 h to obtain purified PCL.

2.3. Synthesis of CL-PU

CL-PU was synthesized with PCL in two steps (Fig. 1). In the first step, a mixture of H_{12} MDI (3.7 g), PCL, and 50 ml THF was introduced into a 500 ml three-necked flask, and afterward reacted at 80 °C for 1 h

Table 1

Molar ratio and Characterization of PCL and CL-PU.

Sample	Feed molar ratio ϵ -CL / H_{12} MDI / 1,4 B.D	Polymer composition ϵ -CL / H_{12} MDI / 1,4 B.D ^a	M_w (kDa) ^b	M_n (kDa) ^b	$\overline{DM}$ ^b
PCL	100 / 0 / 0	—	19.8	7.4	2.6
CL-PU	50 / 37 / 13	69/20/11	62	21.1	2.9

^a Molar percentage determined by 1H NMR.

^b Determined by GPC.

under nitrogen flow then cooled at room temperature. In the second step, 1, 4 BD (1.3 g) and DBTDL (1% relative to the reactant) were added as catalysts and extenders to the flask at room temperature, respectively. The reaction was carried out at 85 °C for 3 h to produce the polymer. Afterward, the reaction product had been cooled to room temperature, then precipitated three times in n-hexane, filtered and cured under vacuum at 40 °C for 24 h.

2.4. Preparation of CL-PU/DCOIT/clay composite

The biodegradable CL-PU/DCOIT/clay composites were prepared by mixing the solution as shown in Fig. 2. CL-PU (75%) was diluted in THF (60 ml) under stirring at 65 °C, while DCOIT (10%) and (15%) clay were introduced into CL-PU solution under magnetic stirrer speed 1200 rpm for 4.5 h. The CL-PU/DCOIT/clay composite solution was finally cast on the substrate. Different CL-PU/DCOIT were synthesized with different percentages of clay, which showed in Table 2.

2.5. Characterization

2.5.1. Proton nuclear magnetic resonance spectroscopy (1H NMR)

1H NMR measurement of PCL and CL-PU were performed by AVANCE Nuclear Magnetic Resonance spectrophotometer (600 MHz, Model DMX-400, USA) instrument.

2.5.2. Fourier transform infrared spectroscopy (FTIR)

The FTIR was carried out by the Nicolet 5700 (FT-IR) (Thermo Fisher, USA) instrument.

2.5.3. Gel permeation chromatography (GPC)

The amount of molecular weight (M_w), average molecular weight (M_n) and molecular weight dispersion ($\overline{DM}$) was calculated by the GPC (Water, USA) system. The sample was dissolved into the THF solvent as eluent with a flow rate of 1 ml/min at 35 °C. The instrument was

calibrated by the polystyrene standard. The GPC result was described in Fig. S1 (Electronic Supporting Information) (ESI†) and Table 1.

2.5.4. Differential scanning calorimetry (DSC)

The crystallization and melting nature of PCL, CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay were evaluated using a differential scanning calorimeter (PE DSC American Perkin-Elmer, USA). Differential scanning calorimetry studies were carried out under nitrogen flow. All samples were heated till the 100 °C with heating rate of 10 °C/min. The melting enthalpy shift (ΔH_m) associated with PCL crystallization was calculated by measuring the region into the melting phase. The crystallinity (Cr) was measured from $Cr = \Delta H_m / W_{PCL} \times \Delta H_f^{\circ} = \Delta H_m / 136.08 \text{ J g}^{-1}$, where ΔH_f° is the change melting enthalpy of PCL with full crystallization [23].

2.5.5. Polarizing optical microscope (POM)

The composite surface morphology was observed using a polarizing microscope with room temperature (ECLIPSE E600W POL, Nikon). The film was prepared by spinning a composite solution in THF (30 wt%) on a spin coater (CHEMAT, KW-4A) at 4200 rpm in the air. The coating glass slide was kept at 40 °C temperature under a vacuum oven for 15 min to remove THF by evaporation to the surface then carefully dried the glass surface [20].

2.5.6. Adhesion test

The adhesion strength of the CL-PU/DCOIT/clay composite coating to the epoxy resin surface has been determined by the adhesion tester (PosiTest AT-A automatic, USA) in accordance with ASTM Standard D4541. The coating composition was constructed using the solution casting method. The data for the adhesion test was done to separate the aluminum dolly (20 mm in diameter) at a rate of 0.2 MPa/s. On average, five different points were evaluated on the surface of each sample [22].

2.5.7. Contact angle (CA) measurements

The analysis of contact angle was performed on a contact angle system (SINDIN Co. Ltd., China) at ambient temperature. The composite coating was prepared using the solution casting method, and 3 μ l of deionized water was applied to the surface of the coating. On average, five different points were measured on the surface of each sample.

2.5.8. Hydrolytic degradation

The hydrolytic degradation test was performed in artificial seawater (ASW), 0.1 and 0.5 mg/ml in ASW containing lipase at room temperature. The composition of the film chip (15 × 15 × 2 mm) was constructed using the solution casting process. The 30% w / v sample was dropped on the panel and kept at room temperature for 3 days to the residual THF. The film chip was weighed (w_1) and submerged in ASW, 0.1 and 0.5 mg in ASW containing lipase container. Every one week, we replaced the earlier water with fresh water. The film chips were removed after a certain time (0, 3, 7, 14, 21, 28 days) from the container, rinsed three times with distilled water, and dried under vacuum at 60 °C till the weight stayed constant. The chip was weighed (w_2) after drying then used the following equation for estimation of the weight loss.

$$\text{Mass loss} = (w_1 - w_2) / w_1 \times 100\% \quad (1)$$

Where w_1 is the weight of sample before immersion and w_2 is the weight of sample after immersion in ASW, 0.1 and 0.5 mg in ASW containing lipase container.

2.5.9. Film thickness

The film thickness of CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay were measured using a digital micrometer screw gauge (Tube Digital Micrometers, Guilin Guanglu Measuring Instrument Co., Ltd, China) [24]. The composition of the film chip (15 × 15 × 2 mm) was constructed using the solution casting same procedure (aforementioned)

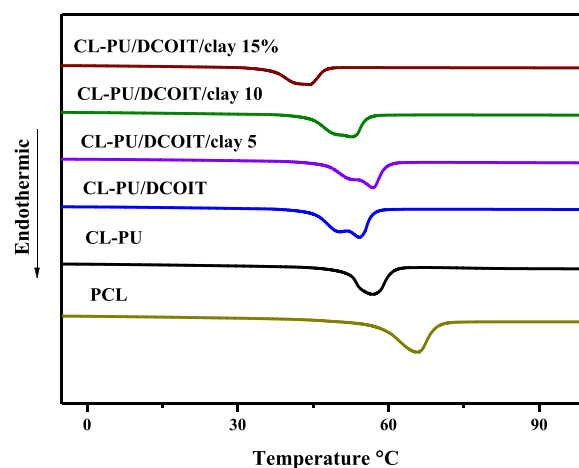


Fig. 3. DSC curve of CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay.

and the film submerged in filtered seawater. Every one week, we replaced the earlier water with fresh seawater. The composite chips were removed after a certain time (0, 7, 14, 21, 28 days) from the container, rinsed three times with distilled water, and dried in the vacuum at 60 °C to a particular time. On average, five different points were measured for each sample.

2.5.10. Diatom adhesion test

Navicula incerta diatom assays were carried out on coating (20 × 20 mm) with epoxy resin panels. Each sample was first soaked in a tank filled with 10 ml of filtered seawater containing diatoms. The culture suspension density of the diatoms were almost 5 μ g/ml and every test was replicated in triplets. The diatoms were then allowed to settle down statically for 24 h at room temperature. After 24 h, the surfaces were quietly washed in filtered seawater to separate the cells that were attached to the surface. Subsequently, chlorophylls were extracted from the *Navicula incerta* diatom with 90% acetone and the chlorophyll concentration was determined by the spectrophotometer UV-vis at 664 nm [22].

2.5.11. Anti-bacterial assay

The anti-microbial action of CL-PU/clay 15, CL-PU/DCOIT5/clay 15, CL-PU/DCOIT10/clay 15 and CL-PU/DCOIT20/clay 15 were evaluated against *E. coli* using the agar diffusion method. The disc of each kind with a weight of 0.15 g was coated on a glass coverslip (China, Glass Round Cover Slip, 0.13 mm thick; 14 mm in diameter) with the use of a spin coater. Bacterial suspensions have been diluted to sterile PBS $\sim 10^8$ CFU/L. The suspension (200 μ l) of the diluted bacterial culture was equally dispersed to the substrate of a Luria broth (LB) agar layer. The CL-PU/clay 15 without DCOIT dispersed with microorganisms was used as a control. The anti-microbial efficiency of samples against bacteria was compared by determining the inhibition region around the samples in the test units after incubation at 37 °C for 24 h [25].

3. Results and discussion

^1H NMR spectroscopy confirmed the chemical composition of the PCL and CL-PU polymers. Fig. 2Sa (ESI†) depicted the ^1H NMR spectrum of PCL in CDCl_3 , 1.35, 1.62 (CH_2) $_3$, 2.28 ($-\text{CH}_2\text{OOC}$), and 4.02 ($-\text{OCCCH}_2-$) [26,27]. Fig. 2Sb (ESI†) depicted the ^1H NMR spectrum of CL-PU in CDCl_3 , 3.39 (δH^j), 1.55 (δH^k), 1.96 (δH^l), 1.70 (δH^m), and 0.95 (δH^n) were found due to H_{12}MDI . The urethane group formation ($-\text{CO}-\text{NH}-$) was detected at 4.81–4.52 (δH^i) [28,29].

The repeating unit of the PCL segment was determined in the CL-PU. The peak area of ($-\text{OCCCH}_2-$) 4.06 ppm was obtained from ^1H NMR, the result was calculated by following equation [30],

Table 2

Molar ratio and characterization of biodegradable CL-PU/DCOIT/clay composite.

Sample	CL-PU / DCOIT / clay ^a	T _m (C) ^b	ΔH _m (J g ⁻¹) ^b	Degree of crystallinity ^b	γ (MPa) ^c	CA ^d
PCL	100	65.4	69.8	51.3%	–	–
CL-PU	100 / 0 / 0	56.8	41.9	30.8%	2.16 ± 0.35	85.4 ± 5.4
CL-PU / DCOIT	90 / 10 / 0	54.3	31.7	23.3%	2.58 ± 0.49	79.5 ± 3.5
CL-PU / DCOIT / clay 5	85 / 10 / 5	56.7	33	24.2%	2.66 ± 0.37	80.8 ± 4.3
CL-PU / DCOIT / clay 10	80 / 10 / 10	52.6	30.8	22.6%	2.93 ± 0.41	74.9 ± 2.1
CL-PU / DCOIT / clay 15	75 / 10 / 15	44.5	23.1	17.0%	3.20 ± 0.23	74.5 ± 1.6

^a Feed molar ratio,^b Determined by DSC,^c Adhesion strength,^d Water contact angle.

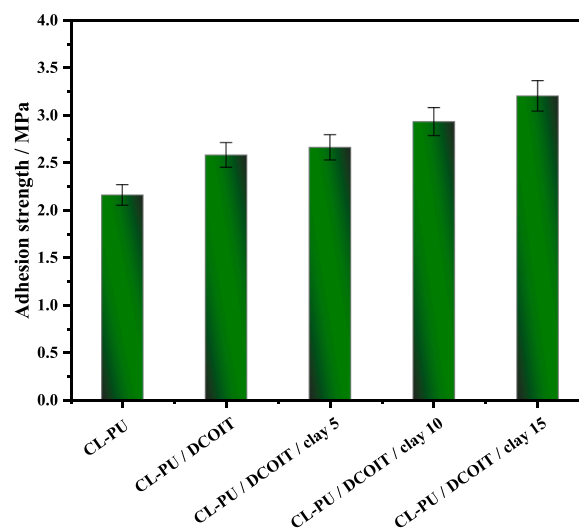
$$DP_{PCL} = (Mn_{PCL}/114) \quad (2)$$

Where the molecular weight of 114 are repeat units of ϵ -CL segments and Mn obtained from GPC (Table 1). There are 127 repeat units of PCL segments in the CL-PU.

The FTIR spectra of PCL and CL-PU are shown in Fig. 3S (ESI[†]). The characteristic band of PCL were 1727 cm⁻¹ (carbonyl stretching), 2800–3000 cm⁻¹ (symmetric and asymmetric CH₂ stretching), 1294 cm⁻¹ (C–O and C–C stretching), 1243 cm⁻¹ (asymmetric C–O–C stretching), and 1167 cm⁻¹ (symmetric C–O stretching) [31]. The spectra of CL-PU at 1729 cm⁻¹ (carbonyl in the 1st amide), 3386 cm⁻¹ (N–H band), 1535 cm⁻¹ (N–H and C–O (2nd amide band) combination band). The absence of the –NCO band at 2275 cm⁻¹ verified that the PU had been cured [32]. These results have shown that CL-PU has been successfully synthesized.

The melting and crystallization behaviour of the samples have been investigated by the DSC experiment. The DSC thermogram of PCL, CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay composite, as shown in Fig. 3 and Table 2. The endothermic peak (T_m) of PCL and CL-PU (control) were observed at 65.4 and 56.8 °C and ΔH_m were 69.8 and 41.91 J g⁻¹. The melting peak of CL-PU/DCOIT was reported at 54.3 °C owing to the DCOIT. The introduction of the clay effect on the degree of crystallization of CL-PU/DCOIT, thus the T_m of CL-PU/DCOIT/clay 15 was 44.5 °C. The percentage (%) of clay was investigated from 5% to 15 wt% (Fig. 3), which indicated that the crystallinity degree of samples had been changed due to the percent amount of clay. The endothermic peak (ΔH_m) of CL-PU/DCOIT and CL-PU/DCOIT/clay 15 were 31.7 J g⁻¹ and 23.15 J g⁻¹ as compared to CL-PU and previous report (T_m = 49.9 °C and ΔH_m = 61.1 J g⁻¹) [20], revealing that the clay changed the crystallinity of CL-PU in the existence of the DCOIT antifoulant.

The spherulite morphology of CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay has been obtained by POM, as shown in Fig. S4 (ESI[†]). CL-PU exhibited well-defined large spherulite morphology, with an average size of 47.6 μm. While the spherulite size for CL-PU/DCOIT composite has been reduced significantly due to the addition of clay. The size of the spherulite changed significantly when the weight percentage of clay increased from 5.0 wt% to 15.0 wt% [33]. The behaviour indicated that the dispersed 5% clay acted as a heterogeneous nucleating agent to facilitate the PCL crystallization in the composite, which was obvious from the rise in the density of the nucleus of crystal and the creation of smaller crystals [34]. This decreased in the size of spherulite was more apparent with a rise in the concentration of clay associated with the results of DSC. As the region of crystalline are thicker than the amorphous areas, which is more complicated for enzyme and water molecules to enter the spherulite because the rate of degradation improves as the size of spherulite reduces [35]. As a result, the rate of

**Fig. 4.** The adhesion strength of CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay.

degradation of composite improved with the increasing of clay.

The detail of adhesion strength is shown in Fig. 4 and Table 2. The adhesion strength of CL-PU had 2.16 ± 0.35 MPa, but the adhesion rose slightly to 2.58 ± 0.49 MPa after the addition of DCOIT. However, the adhesive strength was substantially enhanced due to the incorporation of clay into CL-PU/DCOIT. The adhesion strength for CL-PU/DCOIT/clay 5, CL-PU/DCOIT/clay 10 and CL-PU/DCOIT/clay 15 were 2.66 ± 0.37, 2.93 ± 0.41 and 3.2 ± 0.23 MPa. The incorporation of clay decreased the size of spherulite and crystallinity degree of CL-PU/DCOIT, which contributed to more amorphous region on the CL-PU/DCOIT surface. This facilitated the spreading and contacting of the composite material to the substrate. But on the other side, the slightly larger spherulite permitted the clay to become more evenly dispersed that improved the strength of the CL-PU/DCOIT/clay composite; therefore the adhesion strength of the composite was enhanced.

The static water contact angle (WCAs) study is an efficient technique for the assessment of hydrophilicity and hydrophobicity, which is shown in Fig. S5 and Table 2. Fig. S5 (ESI[†]) depicted that the CL-PU had a contact angle of 85.4 ± 5.4, which showed an enhancement in the low hydrophilicity of the CL-PU reported with PCL efficiency. The contact angle of the CL-PU/DCOIT and CL-PU/DCOIT/clay 5 were 79.5 ± 3.5 and 80.8 ± 4.3, which indicated that the WCAs of CL-PU/DCOIT/clay 5 was slightly higher than CL-PU/DCOIT. However, the WCAs of CL-PU/DCOIT/clay 10 and CL-PU/DCOIT/clay 15 were 74.9 ± 2.1 and 74.5 ± 1.6. The result indicated that the increasing of clay in composite enhanced hydrophilicity and decreased the WCAs of the composite due

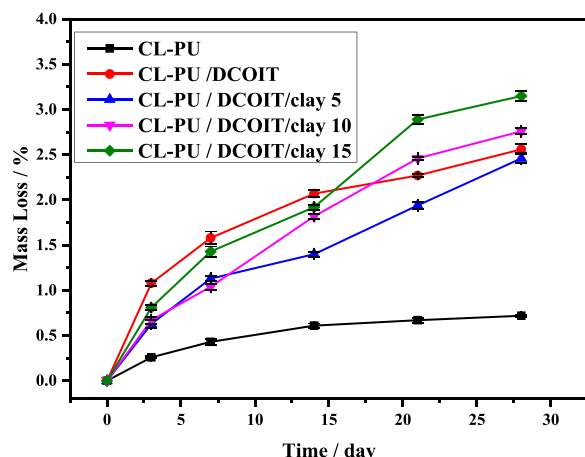


Fig. 5. The time dependence of hydrolytic degradation of CL-PU, CL-PU/clay and CL-PU/ DCOIT/clay in artificial seawater at room temperature.

to reduction in the crystallinity of the PCL.

The hydrolytic degradation efficiency of the CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay chips in the ASW are shown in Fig. 5. Mass loss in all chips began after 3 days. The CL-PU chip resulted in a $0.72 \pm 0.01\%$ weight loss owing to the low rate of degradation due to its considerably higher crystallinity and lower hydrophilicity. The crystallinity of the CL-PU severely hindered the absorption of water, as crystallinity was dominant with almost the entire hard segment. As a result, the water molecules barely penetrated into the polymer matrix and only degraded the surface film. The percentage of mass losses of CL-PU/DCOIT and CL-PU/DCOIT/clay were $2.56 \pm 0.05\%$ and $3.15 \pm 0.06\%$ after 28 days. However, the rate of degradation of CL-PU/DCOIT could be adjusted by dispersing of clay. The rate of degradation of clay was related to the existence of a hydroxyl group on the edge of clay that facilitated rapid hydrolytic degradation of the composite. The substantially greater degradation rate of CL-PU/DCOIT/clay 15 was possible because of the greater connection between CL-PU/DCOIT and clay ensuring from the lowest melting heat. In other words, the increasing of clay decreased the spherulite size and crystallinity of the PCL in the composite, thus the amorphous region and crystallinity in CL-PU/DCOIT/clay were dispersed more uniformly than CL-PU/DCOIT. Therefore the clay frequently allowed water molecules to enter deeply into CL-PU/DCOIT composite, leading to a high hydrolysis rate. The addition of clay into the CL-PU/DCOIT continued to improve composite

surface hydrophilicity, as confirmed by WCAs. While the existence of clay also regulated the slow or higher release rate of the antifoulant, which is considerable for marine anti-fouling. The system has a limited life cycle if antifoulants are released too rapidly. On the other hand, the system cannot prevent biofouling if the antifoulants are released more slowly.

The copolymerization of ϵ -CL with L, lactide [22] or blending with clay [20] unit was a strategy used by our research group or previously reported to reduce crystallinity and improve the biodegradability of degradable polyurethane or PCL polymer. But the previously report revealed that the hydrolytic degradation of blending of PCL/clay/DCOIT composite was lower than PCL/DCOIT in ASW [20]. In our study, CL-PU/DCOIT/clay composite showed better hydrolytic degradation (3.15%) compared to CL-PU/DCOIT and previous report ($<3.0 \text{ mg/cm}^2$) [20] in ASW. Our study demonstrated that CL-PU/DCOIT/clay would be degraded under seawater in order to evaluate the eco-friendly release of antifoulant.

Microbial communities may secrete enzymes that cause composites biodegradation in maritime ecosystems. The enzymatic degradation of CL-PU/DCOIT/clay was also studied in different concentrations of enzyme media (lipase) at room temperature, as shown in Fig. 6. It is notable here that the rate of degradation for CL-PU in ASW without enzyme was relatively slow. Remarkably, just $0.72 \pm 0.01\%$ weight loss was observed after 28 days (aforementioned). On the other hand, the rate of degradation increased considerably in CL-PU/DCOIT/clay and CL-PU/DCOIT composite were observed at the same time, above mentioned, respectively. The existence of enzymes is a prerequisite for the beginning of hydrolysis and subsequently biodegradation. The CL-PU was shown to increase biodegradation by the enzyme secreted by several bacteria [36–38]. The weight loss of CL-PU was $1.1 \pm 0.1\%$ and $2.69 \pm 0.20\%$ at the concentration of 0.1 and 0.5 mg/ml enzymatic solution. The result indicated that enzymatic degradation of CL-PU has been dramatically increased. The enzymatic degradation rate of CL-PU/DCOIT/clay became much quicker than that of CL-PU. CL-PU/DCOIT/clay 15 also showed a greater rate of biodegradation than that of CL-PU/DCOIT in the existence of clay. Because DCOIT molecules were trapped in amorphous regions, but the existence of clay reduced the size of PCL spherulite in CL-PU/DCOIT composite, therefore it made easier to attack by the enzyme. The amorphous portion was first attacked/eaten via an enzyme that created more crystals in the samples. It is reasonable to consider the rate of biodegradation to become more extremely slow for samples with a high amount of crystallinity. Furthermore, it is well known that the activity of enzymes secreted by microorganisms in seawater continues to play an effective role in the

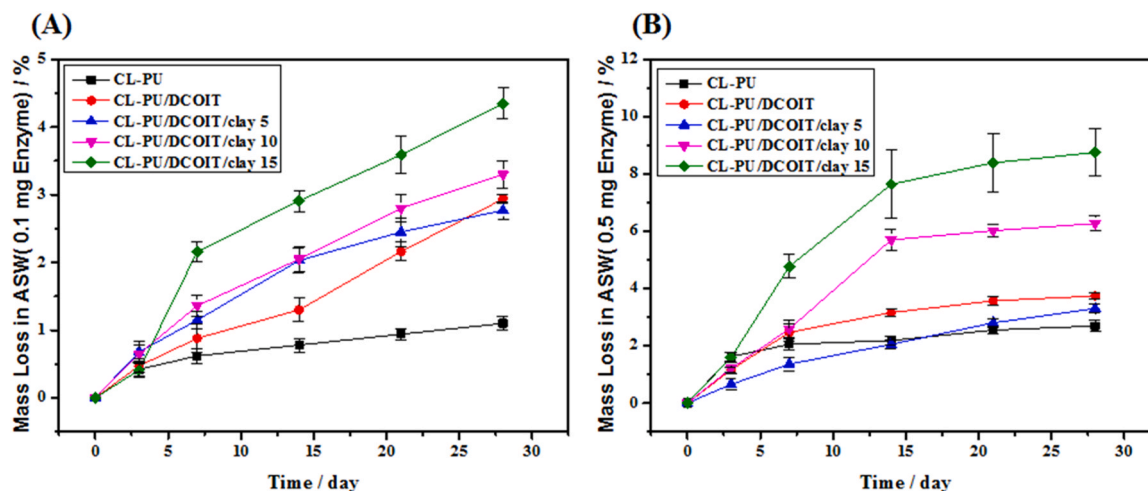


Fig. 6. The time dependence of enzymatic degradation of CL-PU, CL-PU/clay and CL-PU/DCOIT/clay in ASW containing (A) 0.1 mg/ml and (B) 0.5 mg/ml lipase at room temperature.

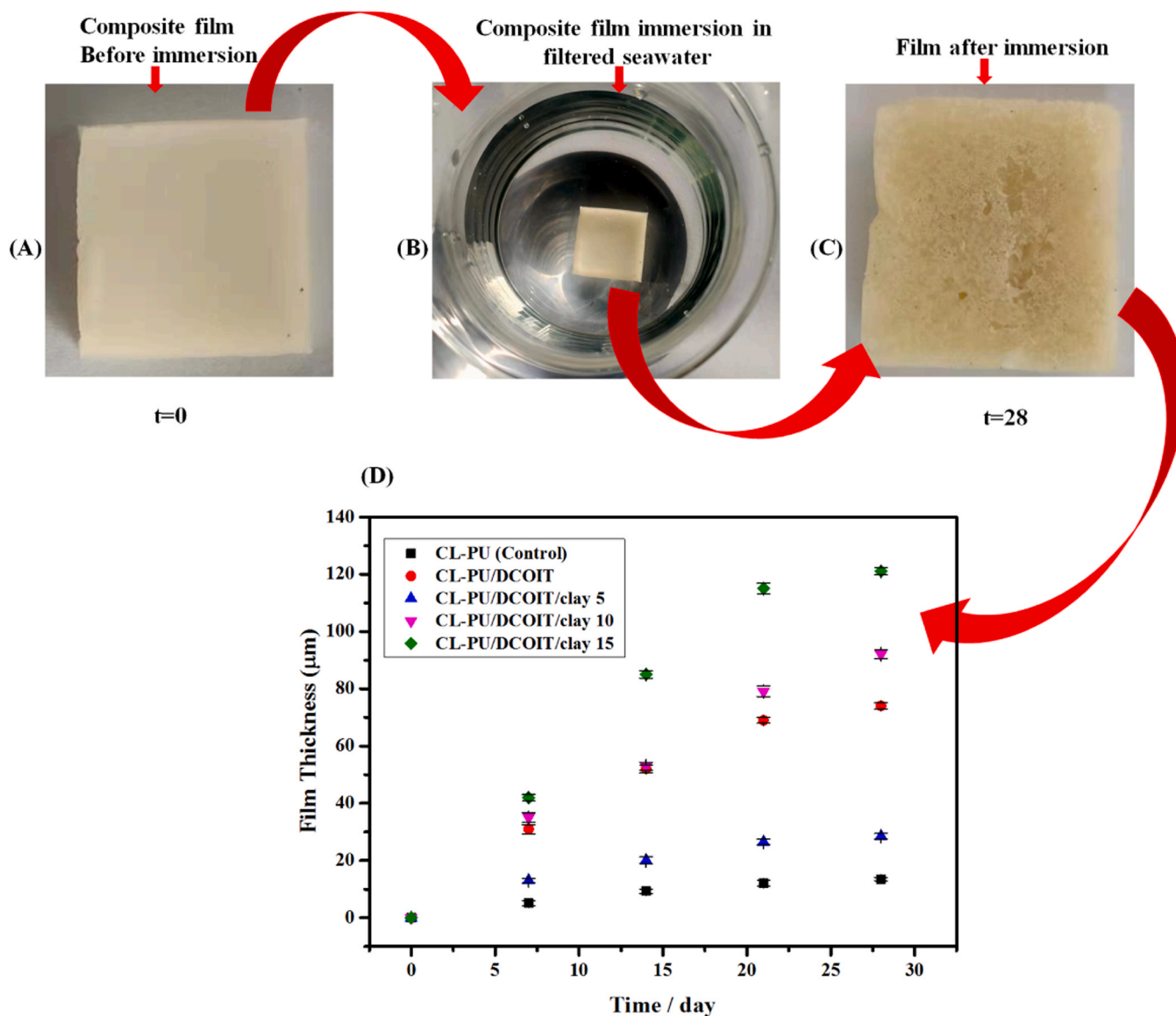


Fig. 7. (A) The image illustrates the prepared composite film before incubation with filtered seawater ($t = 0$), (B) The film chip immersed in static seawater, (C) the effect of immersion on the film at room temperature for 28 days ($t = 28$) and (D) the time dependence of film thickness of CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay in seawater at room temperature.

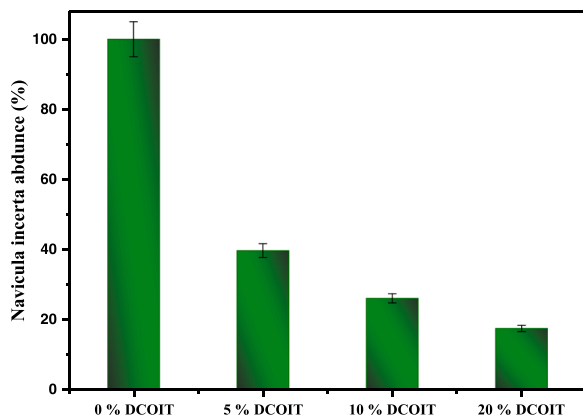


Fig. 8. The abundance of diatoms on the CL-PU/clay composite with different contents of DCOIT.

degradation of the composite. Therefore the mass loss of CL-PU/DCOIT and CL-PU/DCOIT/clay 15 were 2.94 ± 0.06 , 3.74 ± 0.12 , and 4.35 ± 0.23 , $8.75 \pm 0.83\%$ at the concentration of 0.1 and 0.5 mg/ml after 28 days. A significant rise in the rate of biodegradability was noticed for CL-PU/DCOIT/clay composite in 0.1 and 0.5 mg/ml enzymatic solution compared to CL-PU and CL-PU/DCOIT (Fig. 6A & B). The results demonstrated that the degradation of composite rate increases with the concentration of enzymes.

Fig. 7 showed that the change of film thickness for CL-PU, CL-PU/DCOIT and CL-PU/DCOIT/clay composite were function of incubation time in seawater at room temperature. Film thickness of CL-PU reduced $13.5 \pm 0.54 \mu\text{m}$ in 28 days incubation in seawater that was decreased $\sim 0.67\%$, displaying a very high structural stability. Similarly, the thickness of CL-PU/DCOIT decreased by $74 \pm 1.12 \mu\text{m}$, which showed higher thickness reduced by $\sim 3.7\%$ than CL-PU and CL-PU/DCOIT/clay 5 ($28.5 \pm 1.03 \mu\text{m}$). The increasing of clay decreased the spherulite size of PCL resulting in a relatively uniform degradation and also increased the amorphous region in the composite. Therefore, the film thickness of CL-PU/DCOIT/clay 10 and CL-PU/DCOIT/clay 15 were 92.9 ± 1.6

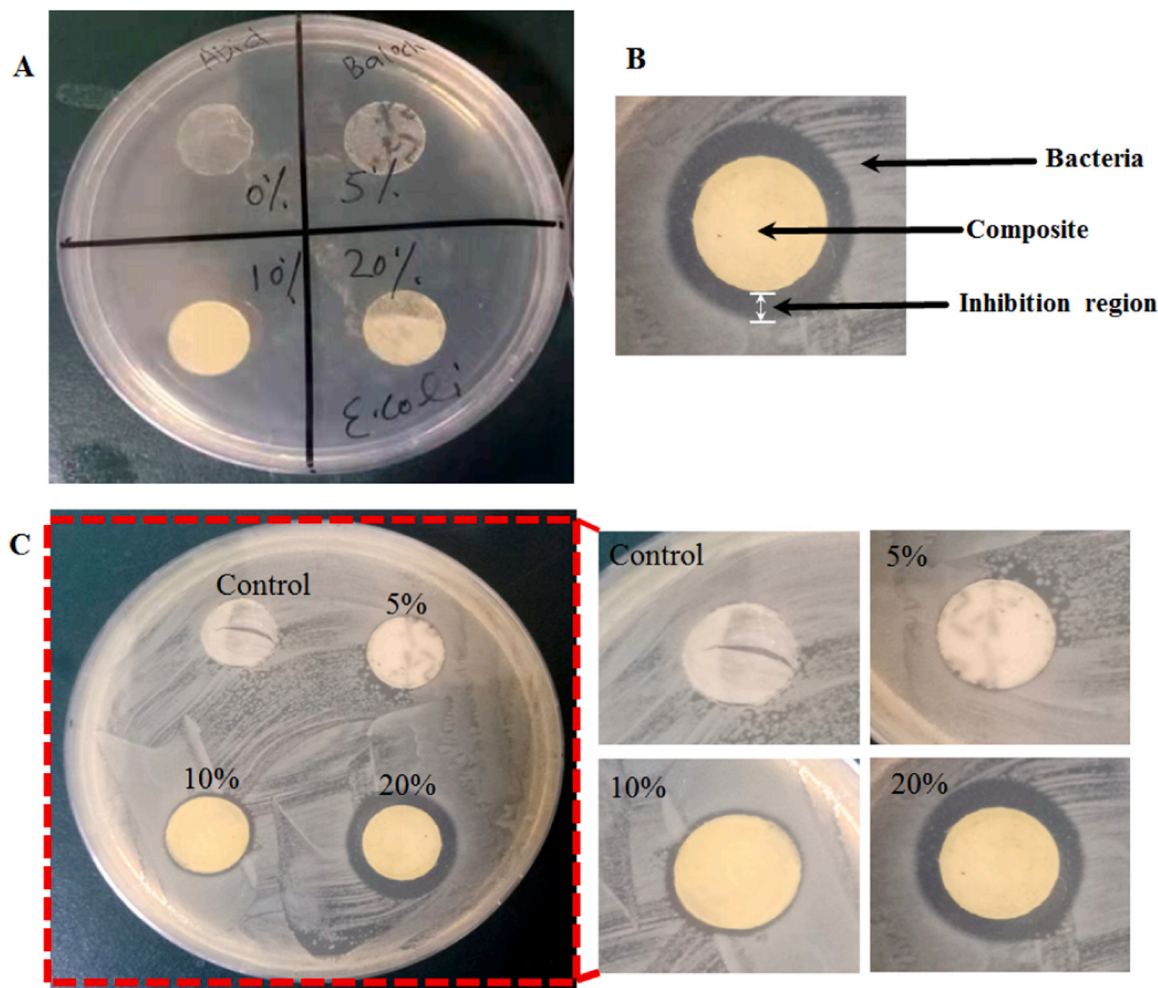


Fig. 9. (A) The digital photograph of before growth of bacteria, (B) Schematic diagram representative of the observed region of inhibition, and (C) illustration of the inhibition region showing the impact of the content activity of DCOIT in the composite against *E. coli*.

(~4.6%) and $121 \pm 1.25 \mu\text{m}$ (~6.05%) indicated a significant decrease. The decrease of the thickness could indicate that the composite surface was eroded; in addition, the thickness is significantly influenced by the degree of erosion and fouling. Therefore, thickness is a vital factor in measuring the biodegradable composite coating system.

The submergence of a surface into the seawater atmosphere can lead to the adhesion of diatom and marine bacteria within 24 h of submergence. The adhered microbes will then form a microbial biofilm that

allows colonized microalgae spores to further facilitate the deposition of macro-organism [39,40]. Therefore anti-fouling is also extremely beneficial for the resistance of the microorganism. The adhesion of *Navicula incerta* diatom cells to the surfaces of the CL-PU/clay/DCOIT was investigated. CL-PU/clay 15 has consisted of 0%, 5%, 10% and 20% of DCOIT. Fig. 8 indicates that the number of *Navicula incerta* diatoms attached to CL-PU / clay 15 after 24 h of incubation. It was found to have a higher quantity of diatoms attached to CL-PU/clay without antifoulant than all of the other samples, indicating that CL-PU/clay without DCOIT was insufficient resistance to diatom adhesion. Consequently, it was observed that the CL-PU/DCOIT20/clay had a lower number of *Navicula incerta* diatoms ($17.5 \pm 1.71\%$), revealing that the DCOIT antifoulant on the surface successfully prevented diatoms. The number of diatom cells was approximately $39.7 \pm 3.40\%$ and $26.1 \pm 2.67\%$ attached to the surface of CL-PU/DCOIT5/clay and CL-PU/DCOIT10/clay, respectively. The anti-diatoms performance has improved only with the content of the DCOIT in the composite.

The anti-bacterial assessment of CL-PU/clay 15, CL-PU/DCOIT5/clay 15, CL-PU/DCOIT10/clay 15 and CL-PU/DCOIT20/clay 15 were measured by the existence of the inhibition region. *E. coli* is a model since it widely exists in the marine environment or daily environment [41]. *E. coli* strain was used to evaluate the effect of composite observing region of inhibition at 37°C for 24 h. The schematic diagram of the inhibition region for the anti-bacterial activity test, as shown in Fig. 9B. The inhibition region is an area around the composite in which there is no formation of the bacterial cells occurs. The higher region is known to

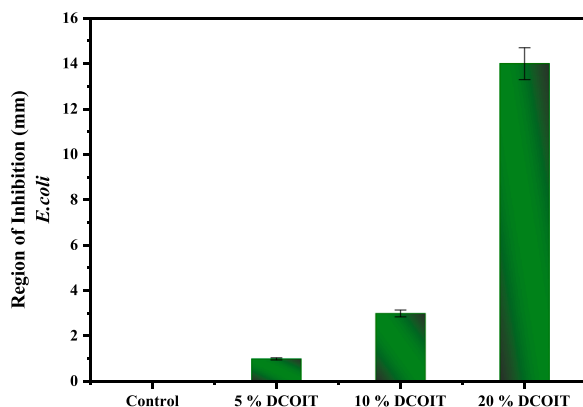


Fig. 10. The inhibition region of bacteria on the CL-PU/clay composite surfaces with different content of DCOIT.

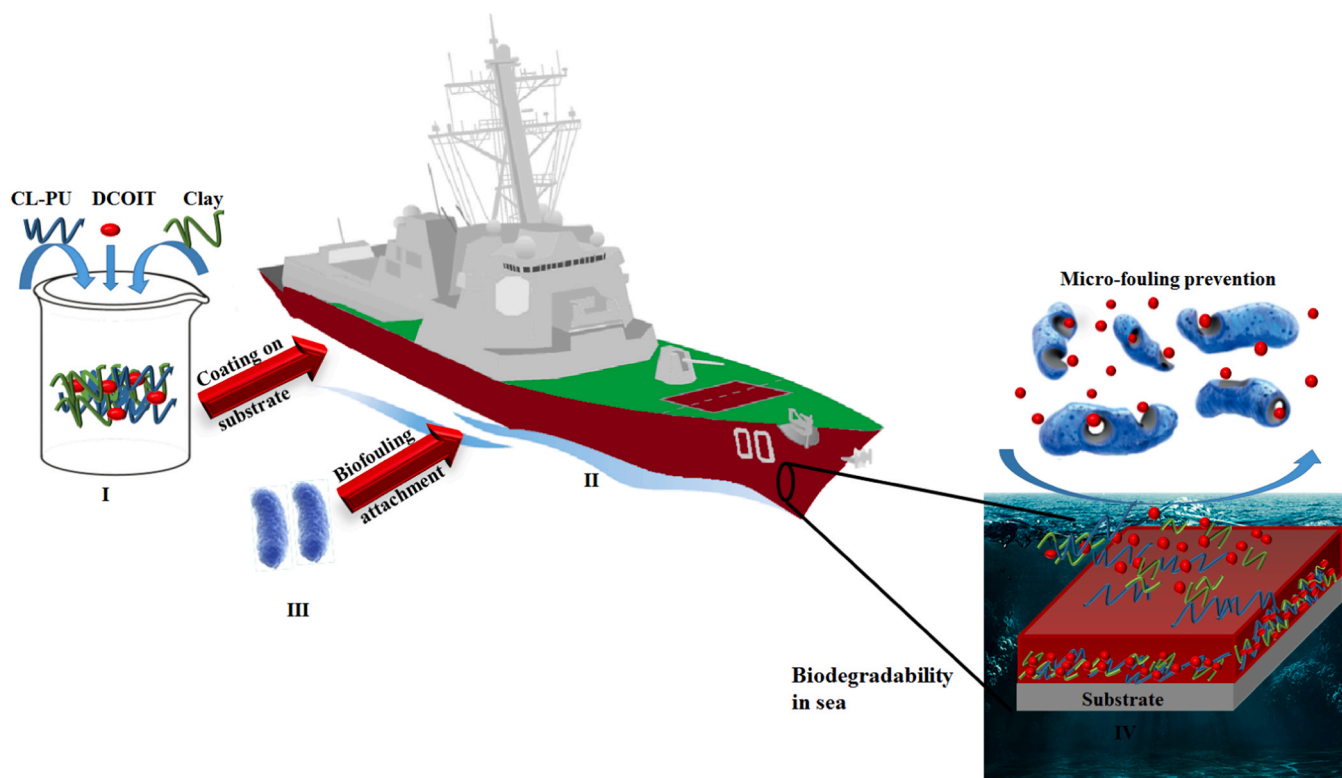


Fig. 11. Anti-fouling properties of composite (I) Schematic illustration for the synthesis of CL-PU/DCOIT/clay composite, (II) the coating casting on the ship's hull, (III) the marine micro-organism attack for the adhesion on the ship's hull and (IV) the mechanism about the release of antifoulants (DCOIT) from the composite during biodegradation and show preventing adhesion of the micro-fouling in the sea.

show a more effective anti-bacterial agent to inhibit bacteria. The image of the inhibition region against *E. coli* for different content of DCOIT with CL-PU/clay 15 is shown in Fig. 9C. Fig. 9C showed that all CL-PU/clay with DCOIT demonstrated some degree of anti-bacterial efficiency and developed an inhibition region around the coverslip of the glass. In comparison, the control showed no inhibition of the formation of the region.

The related ring diameter of the region of inhibition is shown in Fig. 9C. The size of the region of inhibition depends on the composite and microorganism. Fig. 10 showed that CL-PU/clay 15 (control) did not exhibit any anti-bacterial activity; it can be seen that the extracellular polymeric substances (EPS) and high concentration of cells were generated via *E. coli* in the absence of DCOIT. The relative number of bacterial cells attached to the control was defined as 100%, as shown in Fig. 10. To enhance the anti-bacterial activity for CL-PU/clay 15, when DCOIT was added from 5% to 20% in CL-PU/clay 15. Fig. 10 showed that both CL-PU/DCOIT5/clay 15 (1 ± 0.58 mm) and CL-PU/DCOIT10/clay 15 (3.5 ± 0.71 mm) had anti-bacterial efficiency as compared to control. The bactericidal compound DCOIT has shown promising results against *E. coli* in bacterial suspension. The key anti-bacterial properties of DCOIT are expected to drive from its ability to transmit easily into cell walls and to react with essential intracellular sulfur-containing proteins within the cell, leading to impaired cell function [42]. DCOIT allows quick inhibition of metabolism and growth, accompanied by inevitable cell damage, leading to loss of viability [8]. The mechanism of action is known to facilitate a nucleophilic attack by the sulfur atom, accompanied by the cleavage of the S–N bond to create amidodisulfide ring. Cell death is the effect of the formation of free radicals and breakdown of thiol protein [42]. On the other side, the biocide could quickly suppress the essential biological mechanism of bacteria, such as energy generation, respiration and growth. It could be undoubtedly shown that DCOIT concentration affects the inhibition region. The bacterium inhibition region increased to 14 ± 1.52 mm when the DCOIT concentration

enhanced 20% in CL-PU/clay 15. Therefore the introduction of DCOIT in the composite was required in order to inhibit the anti-bacterial action. To compare the results of the inhibition region of bacteria and diatoms on the CL-PU/DCOIT/clay coating as shown in Table S1 (ESI[†]). It is clear that the results of 5% and 10% DCOIT with CL-PU/clay revealed a lower inhibition region of bacteria (1 and 3.5 mm) compared to diatoms (60.3% and 73.9% reduction). In addition, CL-PU/clay with 20% DCOIT exhibited excellent anti-fouling activity against both bacteria (inhibition region 14 mm) and diatom (82.5% reduction) compared to control.

In the published article, the modified CL/clay/DCOIT showed antifouling activity against marine organisms in the seawater [20]. The DCOIT is a highly hydrophobic compound with poor water solubility and biodegradability. DCOIT was awarded the first Green Chemistry Environmental protection agency (EPA) award in 1996 for TBT substitution in anti-fouling materials and has been widely used in the anti-fouling techniques [15]. With our method, we seek to resist the initial phase of biofilm formation, in which everything begins with the adherence of microbes to surfaces. CL-PU/DCOIT/clay 15 with good anti-bacterial efficiency is broad potential for biofilm-resistant coating.

3.1. Proposed mechanism

According to the above result, we can propose a potential mechanism. This CL-PU combined with the antifoulant (DCOIT) and the higher reduction of spherulite size by clay. The anti-fouling activity of DCOIT can prevent the adhesion of biofouling. In the marine environment, the hydrolysis degradation of the polymer in the composite coating can include the cleavage of the ester bond in the PCL and urethane segment (NHCOO). Fungus secretes enzyme with alkaline in the marine environment; therefore, cleavage of the durable composite can occur due to different enzymes (Lipase, Protease, etc) with alkaline. The DCOIT is physically incorporated into a biodegradable composite. Moreover, in an aqueous environment, the content of clay can decrease the size of

spherulite and crystallinity of the PCL in the composite. Thus, the contents of clay can also regulate the slow or higher release rate of the DCOIT for the prevention of the biofouling attachment on the ship's hull (Fig. 11). This proposed mechanism might be the aim of the excellent performance of the composite in the complex marine environment due to enzymatic and hydrolytic degradation.

4. Conclusion

We have prepared CL-PU synthesized by ring open polymerization and condensation reaction, then prepared CL-PU further modified with DCOIT and clay by mixing of the solution to make a composite. Our study showed that the addition of clay increased the degradation rate of CL-PU/DCOIT and improved the amorphous interfacial region in the composite with an amount that reduced the crystallinity of PCL. The composite was used as a carrier and release system for DCOIT. Moreover, the results of anti-diatom (*Navicula incerta*) and anti-bacterial (*E. coli*) activity indicated that the CL-PU/DCOIT20/clay composite had excellent anti-fouling ability than other composites. These results indicated that the biodegradable CL-PU/DCOIT/clay composite would have remarkable durability and excellent anti-fouling activity for marine biofouling.

CRediT authorship contribution statement

Abid Ali: Experiment, Investigation, Conceptualization, Methodology, Formal analysis, Writing - original draft. **Yue Xiao:** Conceptualization. **Lina Song:** Validation. **Qingqing Rao:** Investigation. **Jiankun Hu:** Conceptualization. **Muhammad Shoaib:** Software. **Bilal Ul Amin:** Review and Editing. **Xiaoli Zhan:** Supervision, Funding acquisition. **Qinghua Zhang:** Supervision, Funding acquisition. Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (No. 21878267) and Natural Science Foundation of Jiangxi (No. 20181BAB206022).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.colsurfa.2021.126946](https://doi.org/10.1016/j.colsurfa.2021.126946).

References

- [1] A. Ali, M.I. Jamil, J. Jiang, M. Shoaib, B.U. Amin, S. Luo, X. Zhan, F. Chen, Q. Zhang, An overview of controlled-biocide-release coating based on polymer resin for marine antifouling applications, *J. Polym. Res.* 27 (2020) 1–17, <https://doi.org/10.1007/s10965-020-02054-z>.
- [2] L.D. Chambers, K.R. Stokes, F.C. Walsh, R.J.K. Wood, Modern approaches to marine antifouling coatings, *Surf. Coat. Technol.* 201 (2006) 3642–3652, <https://doi.org/10.1016/j.surfcoat.2006.08.129>.
- [3] Y.K. Demirel, M. Khorasanchi, O. Turan, A. Incecik, On the importance of antifouling coatings regarding ship resistance and powering, *Low. Carbon Shipp Conf.* 44 (2013) 1–13, <https://doi.org/10.1037/0033-3204.44.3.0>.
- [4] M.P. Schultz, J.A. Bendick, E.R. Holm, W.M. Hertel, Economic impact of biofouling on a naval surface ship, *Biofouling* 27 (2011) 87–98, <https://doi.org/10.1080/08927014.2010.542809>.
- [5] K. Yebra, D. Meseguer, K. Søren, Dam-Johansen, Antifouling technology—past, present and future steps towards efficient and environmentally friendly antifouling coatings, *Prog. Org. Coat.* 50 (2004) 75–104, <https://doi.org/10.1016/j.porgcoat.2003.06.001>.
- [6] E. Almeida, T.C. Diamantino, O. de Sousa, Marine paints: the particular case of antifouling paints, *Prog. Org. Coat.* 59 (2007) 2–20, <https://doi.org/10.1016/j.porgcoat.2007.01.017>.
- [7] IMO, Focus on IMO: anti-fouling systems, *Int. Marit. Organ* 44 (2002) 1–31, <https://doi.org/10.1097/TA.0b013e31817de3f4>.
- [8] T.M. Williams, The mechanism of action of isothiazolone biocides, *Power Plant Chem.* 9 (2007) 14–22.
- [9] J. Morley, A.J. Olivera, M.H. Charlton, Theoretical studies on the biocidal activity of 5-chloro-3-isothiazolone, *Theochem. J. Mol. Struct.* 280 (1998) 0–7, [https://doi.org/10.1016/S0166-1280\(97\)00342-4](https://doi.org/10.1016/S0166-1280(97)00342-4).
- [10] A. Khalaj, N. Adibpour, A.R. Shahverdi, M. Daneshmand, Synthesis and antibacterial activity, *Eur. J. Med. Chem.* 39 (2004) 699–705, <https://doi.org/10.1016/j.ejmech.2004.04.004>.
- [11] C. Huang, S. Hsieh, P. Kuo, W. Jane, J. Tu, Y. Wang, C. Ko, Impact of disinfectant and nutrient concentration on growth and biofilm formation for a *Pseudomonas* strain and the mixed cultures from a fine papermachine system, *Int. Biodeterior. Biodegrad.* 63 (2009) 998–1007, <https://doi.org/10.1016/j.ibiod.2009.07.004>.
- [12] J.S. Chapman, Biocide resistance mechanisms, *Int. Biodeterior. Biodegrad.* 51 (2003) 133–138, [https://doi.org/10.1016/S0964-8305\(02\)00097-5](https://doi.org/10.1016/S0964-8305(02)00097-5).
- [13] R.J.C.A. Steen, F. Ariese, B. Van Hattum, J. Jacobsen, A. Jacobson, Monitoring and evaluation of the environmental dissipation of the marine antifoulant (DCOIT) in a Danish Harbor, *Chemosphere* 57 (2004) 513–521, <https://doi.org/10.1016/j.chemosphere.2004.06.043>.
- [14] M. Moradi, J. Duan, X. Du, Investigation of the effect of 4, 5-dichloro-2-n-octyl-4-isothiazolin-3-one inhibition on the corrosion of carbon steel in *Bacillus* sp. inoculated artificial seawater, *Corros. Sci.* 69 (2013) 338–345, <https://doi.org/10.1016/j.corsci.2012.12.017>.
- [15] G. Dai, Q. Xie, S. Chen, C. Ma, G. Zhang, Biodegradable poly (ester) -poly (methyl methacrylate) copolymer for marine, *Prog. Org. Coat.* 124 (2018) 55–60, <https://doi.org/10.1016/j.porgcoat.2018.08.003>.
- [16] S. Chen, C. Ma, G. Zhang, Biodegradable polymers for marine antibiofouling: poly (ε-caprolactone)/poly(butylene succinate) blend as controlled release system of organic antifoulant, *Polym. (Guildf.)* 90 (2016) 215–221, <https://doi.org/10.1016/j.polymer.2016.03.017>.
- [17] J. Pan, Q. Xie, H. Chiang, Q. Peng, P.Y. Qian, C. Ma, G. Zhang, "From the nature for the nature": An eco-friendly antifouling coating consisting of poly(lactic acid)-based polyurethane and natural antifoulant, *ACS Sustain. Chem. Eng.* 8 (2020) 1671–1678, <https://doi.org/10.1021/acssuschemeng.9b06917>.
- [18] F. Fay, I. Linossier, E. Renard, K. Valle, Development of poly(ε-caprolactone-co-L-lactide) and poly(ε-caprolactone-co-D-valerolactone) as new degradable binder used for antifouling paint, *Eur. Polym. J.* 43 (2007) 4800–4813, <https://doi.org/10.1016/j.eurpolymj.2007.07.045>.
- [19] C. Ma, L. Xu, W. Xu, G. Zhang, Degradable polyurethane for marine anti-biofouling, *J. Mater. Chem. B* 1 (2013) 3099–3106, <https://doi.org/10.1039/c3tb20454e>.
- [20] J. Yao, S. Chen, C. Ma, G. Zhang, Marine anti-biofouling system with poly (ε-caprolactone)/clay composite as carrier of organic antifoulant, *J. Mater. Chem. B* 2 (2014) 5100–5106, <https://doi.org/10.1039/c4tb00545g>.
- [21] ASTM D1141, Standard Practice for the Preparation of Substitute Ocean Water, 2013, <https://doi.org/10.1520/D1141-98R13.2>.
- [22] A. Ali, L. Song, J. Hu, J. Jiang, Q. Rao, M. Shoaib, S. Fahad, Y. Cai, X. Zhan, F. Chen, Q. Zhang, Synthesis and characterization of caprolactone based polyurethane with degradable and antifouling performance, *Chin. J. Chem. Eng.* (2020), <https://doi.org/10.1016/j.cjche.2020.11.007>.
- [23] J. Ma, C. Ma, G. Zhang, Degradable polymer with protein resistance in a marine environment, *Langmuir* 31 (2015) 6471–6478, <https://doi.org/10.1021/acs.langmuir.5b01720>.
- [24] Y. Alparslan, T. Baygar, Effect of chitosan film coating combined with orange peel essential oil on the shelf life of deepwater pink shrimp, *Food Bioprocess Technol.* 10 (2017) 842–853, <https://doi.org/10.1007/s11947-017-1862-y>.
- [25] X. Chen, S. Ku, J.A. Weibel, E. Ximenes, X. Liu, M.R. Ladisch, S.V. Garimella, Enhanced antimicrobial efficacy of bimetallic porous CuO microspheres decorated with Ag nanoparticles, *ACS Appl. Mater. Interfaces* 9 (2017) 39165–39173, <https://doi.org/10.1021/acsami.7b11364>.
- [26] D. Psimadas, P. Georgoulas, V. Valotassiou, G. Loudos, Thermosensitive PEG–PCL–PEG hydrogel controlled drug delivery system: sol–gel–sol transition and in vitro drug release study, *J. Pharm. Sci.* 101 (2012) 2271–2280, <https://doi.org/10.1002/jps.21694>.
- [27] R. Mahou, C. Wandrey, Versatile route to synthesize heterobifunctional poly (ethylene glycol) of variable functionality for subsequent pegylation, *Polym. (Basel)* 4 (2012) 561–589, <https://doi.org/10.3390/polym4010561>.
- [28] J. Cui, K. Kratz, M. Heuchel, B. Hiebl, A. Lendlein, Mechanically active scaffolds from radio-opaque shape-memory polymer-based composites, *Polym. Adv. Technol.* 22 (2011) 180–189, <https://doi.org/10.1002/pat.1733>.
- [29] T.Y. Shih, J.D. Yang, H.W. Jia, J.H. Chen, Synthesis and properties of biodegradable segmented poly-ε-caprolactone, *J. Med. Biol. Eng.* 34 (2014) 238–242, <https://doi.org/10.5405/jmbe.1534>.
- [30] M.H. Huang, S. Li, J. Coudane, M. Vert, Synthesis and characterization of block copolymers of ε-caprolactone and DL-lactide initiated by ethylene glycol or poly (ethylene glycol), *Macromol. Chem. Phys.* 204 (2003) 1994–2001, <https://doi.org/10.1002/macp.200350054>.
- [31] M. Azizi, M. Azimzadeh, M. Afzali, M. Alafzadeh, S.H. Mirhosseini, Characterization and optimization of using *calendula officinalis* extract in the fabrication of polycaprolactone/gelatin electrospun nanofibers for wound dressing applications, *J. Adv. Mater. Process* 6 (2018) 34–46, https://doi.org/http://jmatpro.iaun.ac.ir/article_623136_24173abe59b9472fd4140377fae51745.pdf.

- [32] P. Król, Ł. Uram, B. Król, K. Pielichowska, M. Walczak, Study of chemical, physico-mechanical and biological properties of 4,4'-methylenebis(cyclohexyl isocyanate)-based polyurethane films, *Mater. Sci. Eng. C* 93 (2018) 483–494, <https://doi.org/10.1016/j.msec.2018.07.082>.
- [33] T. Wu, T. Xie, G. Yang, Applied Clay Science Preparation and characterization of poly (ϵ -caprolactone)/Na⁺-MMT nanocomposites, *Appl. Clay Sci.* 45 (2009) 105–110, <https://doi.org/10.1016/j.clay.2009.02.009>.
- [34] P. Maiti, P.H. Nam, M. Okamoto, N. Hasegawa, A. Usuki, Influence of crystallization on intercalation, morphology, and mechanical properties of polypropylene/clay nanocomposites, *Macromolecules* 35 (2002) 2042–2049, <https://doi.org/10.1021/ma010852z>.
- [35] Y. Hou, J. Chen, P. Sun, Z. Gan, G. Zhang, In situ investigations on enzymatic degradation of poly (3 -caprolactone), *Polym. (Guildf.)* 48 (2007) 6348–6353, <https://doi.org/10.1016/j.polymer.2007.08.054>.
- [36] N.K. Singh, B. Das Purkayastha, J.K. Roy, R.M. Banik, M. Yashpal, G. Singh, S. Malik, P. Maiti, Nanoparticle-induced controlled biodegradation and its mechanism in poly(ϵ -caprolactone), *ACS Appl. Mater. Interfaces* 2 (2010) 69–81, <https://doi.org/10.1021/am900584r>.
- [37] C.V. Benedict, W.J. Cook, P. Jarrett, J.A. Cameron, S.J. Huangt, J.P. Belli, Fungal degradation of polycaprolactones, *J. Appl. Polym. Sci.* 28 (1983) 327–334, <https://doi.org/10.1002/app.1983.070280128>.
- [38] H. Nishida, S. Suzuki, Y. Tokiwa, Distribution of poly(β -propiolactone) aerobic degrading microorganisms in different environments, *J. Environ. Polym. Degrad.* 6 (1998) 43–58, <https://doi.org/10.1023/A:1022878512743>.
- [39] M.S. Selim, S.A. El-Safty, M.A. Shenashen, S.A. Higazy, A. Elmarakbi, Progress in biomimetic leverages for marine antifouling using nanocomposite coatings, *J. Mater. Chem. B* 8 (2020) 13701–13732, <https://doi.org/10.1039/c9tb02119a>.
- [40] J.A. Callow, M.E. Callow, Trends in the development of environmentally friendly fouling-resistant marine coatings, *Nat. Commun.* 2 (2011) 210–244, <https://doi.org/10.1038/ncomms1251>.
- [41] R. Chen, Y. Zhang, Q. Xie, Z. Chen, C. Ma, G. Zhang, Transparent polymer-ceramic hybrid antifouling coating with superior mechanical properties, *Adv. Funct. Mater.* 2011145 (2021) 1–7, <https://doi.org/10.1002/adfm.202011145>.
- [42] A.M. El-Shamy, T.Y. Soror, H.A. El-Dahan, E.A. Ghazy, A.F. Eweas, Microbial corrosion inhibition of mild steel in salty water environment, *Mater. Chem. Phys.* 114 (2009) 156–159, <https://doi.org/10.1016/j.matchemphys.2008.09.003>.